

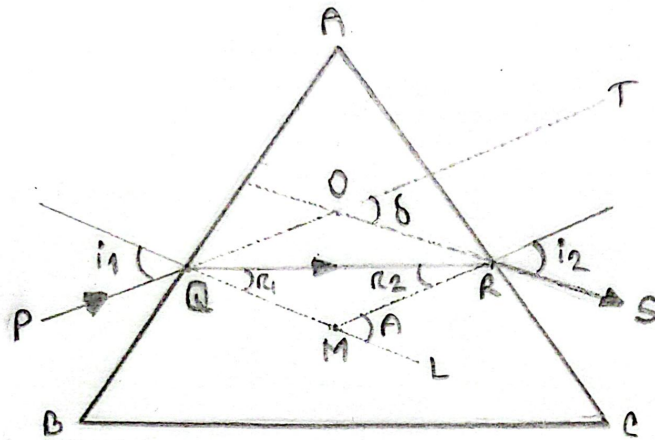


Figure 1: Refraction mechanism through the prism

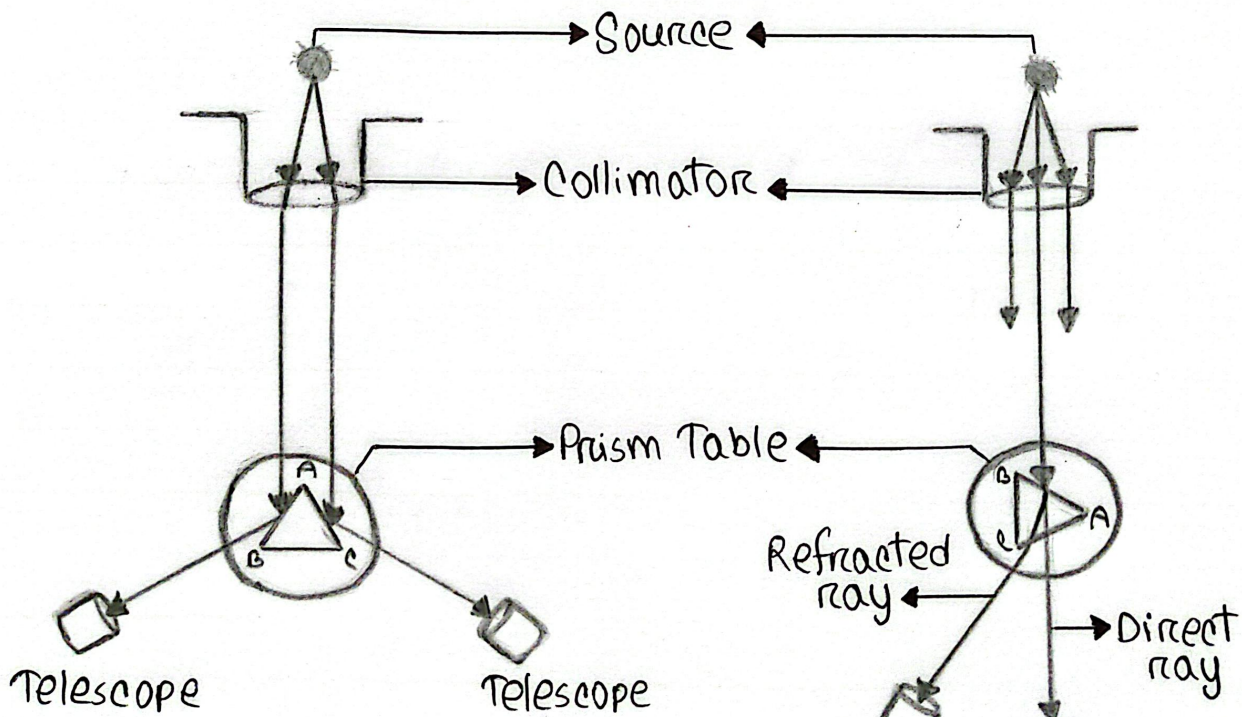


Figure 2: Measurement of the prism angle, A

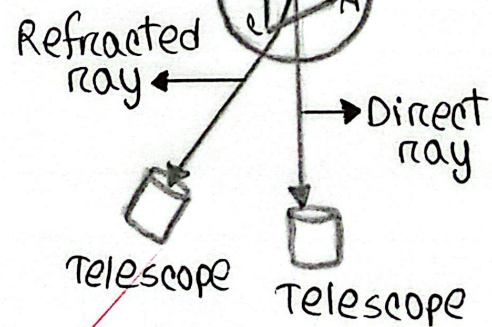


Figure 3: Measurement of the angle of minimum deviation, δ_m

EXPT. NO. 02

DATE: 07.12.2025

EXPERIMENT OF Determination of the refractive index of the material of a prism by a spectrometer

Theory:

Refractive index is the ratio of the speed of light in vacuum to the speed of light in a medium. Let, a ray PQ be the incident on the first face AB of a prism and then after passing through the prism, finally emerge out through the other face AC in the direction RS. Let i_1 and r_1 be the respective angles of incidence and refraction at the first face of the prism and i_2 and r_2 be the corresponding quantities for the second face. Now, the deviation of the ray is given by the angle $\angle SOT$,

$$\delta = (i_1 - r_1) + (i_2 - r_2) \dots (i)$$

But, in the minimum deviation position, the ray passes symmetrically through the prism so that $i_1 = i_2$ and $r_1 = r_2$. Therefore, the angle of minimum deviation,

$$\delta_m = 2(i_1 - r_1) \dots (ii)$$

But,

$$\angle LMR = r_1 + r_2 = 2r_1 = A \dots (iii)$$

i.e.

$$r_1 = A/2$$

Where, A = Angle of prism

EXPT. NO. EXPERIMENT OF

DATE :

From eq (ii) and (iii), we get, $i_1 = \frac{A + \delta_m}{2}$

$$\text{Hence, } \mu = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$$

Required Apparatus:

- (i) Spectrometer
- (ii) Prism
- (iii) Spirit level
- (iv) Sodium (Na) light

Procedure:

- (i) The telescope, collimator and prism table were leveled and aligned, the slit and crosswires were focused for clear observations
- (ii) The vernier constants (vc) of both verniers were determined and the prism was positioned so its vertical plane aligned with the collimator axis
- (iii) Parallel rays were directed onto prism faces AB and AC, and the resulting images were observed through the telescope. Vernier readings were recorded for both direct and reflected rays. For each observation, readings were taken for both

EXPT. NO.

EXPERIMENT OF

DATE :

scales (A and B) for the left and right position of the image. Values such as MSR, VSD, VSR and TR were calculated for each scale

(iv) The prism angle A was calculated separately for scale A and scale B using $A = \frac{P-Q}{2}$, where P and Q were the total readings of the left and right positions. The mean of two scale values were taken as a final prism angle

(v) The prism was positioned on the table to measure the angle of minimum deviation δ_m . Then the prism was gradually rotated until the image approached the direct path of the rays and then began moving back, indicating the point of minimum deviation

(vi) The telescope was focused on the image at minimum deviation and vernier readings were recorded for both scales. The difference $P-Q$ was calculated for each scale and the mean provided the final minimum deviation δ_m

(vii) Direct light was observed without the prism and vernier readings were recorded. Using the measured prism angle A and minimum deviation δ_m , the refractive index μ of the prism material was calculated

Calculations:

$$\text{Refractive index, } \mu = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$$

$$= 1.51$$

$$\begin{aligned} \text{Error} &= \left| \frac{1.51 - 1.5}{1.5} \right| \times 100 \% \\ &= 0.67 \% \end{aligned}$$

EXPT. NO. EXPERIMENT OF

DATE :

Experimental Data:

Table 1: Data for angle of prism (A)

No. of obs	Scale	Readings of the reflected images								A	Mean
		LEFT				RIGHT				=	A
		MSR	VSD	VSR	TR	MSR	VSD	VSR	TR	$\frac{P+Q}{2}$	(deg)
		M (deg)	D (div)	$V=D \times VC$ (deg)	$P=M+V$ (deg)	M (deg)	D (div)	$V=D \times VC$ (deg)	$Q=M+V$ (deg)	(deg)	
1	A	331.5	14	0.233	331.733	209	20	0.33	209.33	61.245	61.276
	B	152	2	0.033	152.033	29	20	0.33	29.33	61.3515	

Table 2: Data for angle of minimum deviation (δ_m)

Z.O.	S.O.	Reading at minimum deviation position				Direct Reading				δ_m = $P-Q$ (deg)	Mean δ_m (deg)
		MSR	VSD	VSR	TR	MSR	VSD	VSR	TR		
		M	D	$V=D \times VC$	$P=M+V$	M	D	$V=D \times VC$	$Q=M+V$		
		(deg)	(div)	(deg)	(deg)	(deg)	(div)	(deg)	(deg)		
1	A	130	16	0.267	130.267	169.5	6	0.1	169.6	39.33	39.42
	B	309.5	1	0.0167	309.5167	349	2	0.033	349.033	39.51	

Results:

$$\mu = 1.51$$

$$\text{ERROR} = 0.67\%$$

EXPT. NO. EXPERIMENT OF

DATE :

Discussion:

In this experiment, the refractive index (μ) of the material of a prism was determined by using a spectrometer. The ideal refractive index of the prism was 1.5 and the experimental value was calculated to be 1.51. This small amount of deviation was noticed due to measurement errors and experimental inconsistencies.

Precautions:

- The source was kept in front of the collimator to make the image bright.
- The crosswire was made coincident with the same edge of the slit image.
- Parallax error was avoided.
- The telescope was rotated in the same direction to avoid backlash error.
- The width of the slit image was made as narrow as possible.